$$p_X(x) = \frac{1}{2} b\, e^{-b|x|} \qquad \int_{-\infty}^{\infty} p_X(x)\,dx = 2\int_0^{\infty} p_X(x)\,dx$$

$$\int \frac{1}{2} b\, e^{-bx}\,dx = \frac{1}{2} b \int e^{-bx}\,dx = -\frac{1}{2} b \frac{e^{-bx}}{b} = -\frac{1}{2} e^{-bx} + C$$

$a = b \leftarrow$ dla wygodnej

$$\int x p_X(x)\,dx = \frac{a}{2}\int x e^{-ax}\,dx = \frac{a}{2}\left(-\frac{1}{a} x e^{-ax} + \frac{1}{a}\int e^{-ax}\,dx\right) + C =$$

$$\frac{a}{2}\left(-\frac{1}{a} x e^{-ax} - \frac{1}{a^2} e^{-ax}\right) + C = -\frac{1}{2} e^{-ax}\left(x + \frac{1}{a}\right) + C$$

$$m_X = 2\int_0^{\infty} x p_X(x)\,dx = 2\left(0 - \left(-\frac{1}{2a}\right)\right) = \frac{1}{a}$$

$$\sigma_X^2 = 2\int_0^{\infty} (x - m_X)^2 p_X(x)\,dx \qquad y = e^{-ax}, \quad \int y\,dy = -\frac{1}{a} e^{-ax}$$

$$(x - m_X)^2 p_X(x) = \left(x^2 - \frac{2}{a} x + \frac{1}{a^2}\right)\left(\frac{1}{2} a y\right) = \frac{1}{2}\left(x^2 a y - 2 x a y + \frac{1}{a} y\right)$$

$$a \int x^2 e^{-ax}\,dx = a\left(\frac{1}{a} x^2 e^{-ax} - \frac{2}{a} x e^{-ax} - \frac{2}{a^2} e^{-ax}\right)$$

$$e^{-ax}\left(x^2 - 2x - \frac{2}{a}\right)$$

$$\int x^2 a e^{-ax} - 2x a e^{-ax} + \frac{1}{a} e^{-ax}\,dx = \left(x^2 y - 2xy - \frac{2}{a} y\right) - 2\left(yx - \frac{1}{a} y\right) - \frac{y}{a^2}$$

$$x^2 y - y = x^2 e^{-ax} - \frac{e^{-ax}}{a^2}$$

$$(x - m_X)^n = \frac{n-1}{a}\left(\frac{n-2}{a}\left(\frac{n-3}{a} e^{-ax}\right)\right)$$

$$\mu_X^n = \frac{(n-1)!}{a^n}$$

$$G_X^2 = x^2 e^{-ax} - \frac{e^{-ax}}{a^2}\Big|_0^{\infty} = \frac{1}{a^2}$$

$$G_X^n = \frac{1}{a^n}$$

$$\gamma_X = \frac{2^n}{a^3} \cdot \frac{a^3}{1}$$

$x > 0$ strona

$$k_X = (4-1)! - 3 = 3 \qquad \gamma_X = 2 - 2 = 0$$